

Simple SECR workflow

Introduction

Centre for Research into Ecological & Environmental Modelling
University of St Andrews

Welcome

A standardized workflow for designing and analyzing camera trap surveys of large cats with spatially explicit capture-recapture (SCR).

- Session 1: Concepts
- Session 2: Simple workflow
- Session 3: Adding complexity
- Session 4: Automatic checks of data integrity
- Session 5: Wrap up

Welcome

A standardized workflow for designing and analyzing camera trap surveys of large cats with spatially explicit capture-recapture (SCR).

- Session 1: Concepts
- Session 2: Simple workflow
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- Session 5: Wrap up

What we expect from participants:

- Some knowledge on statistics
- Know how to work with R and RStudio

Ready for the practicals

Data:

1. Download from XXXXXXXXXX
2. Unzip into dedicated secr workshop folder

Software required:

1. R: <https://cran.rstudio.com/>
2. RStudio: <https://posit.co/download/rstudio-desktop>
3. R packages (and dependencies):

- | | |
|-------------|-------------|
| ▪ dplyr | ▪ patchwork |
| ▪ elevatr | ▪ secr |
| ▪ gdistance | ▪ sf |
| ▪ ggplot2 | ▪ stringr |
| ▪ lubridate | ▪ tidyr |
| ▪ MASS | ▪ terra |

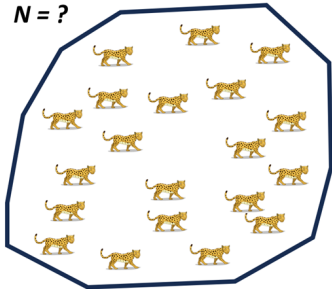
Session 1 objectives

Main goal: Understand the concepts of SCR surveys and be ready for the practicals

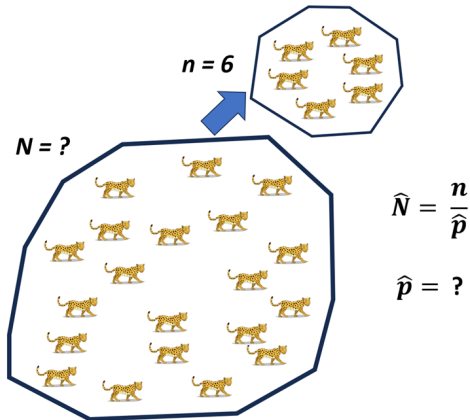
1. Ecological surveys and sampling
2. Capture-recapture surveys
3. Spatial capture-recapture surveys

Population and sample

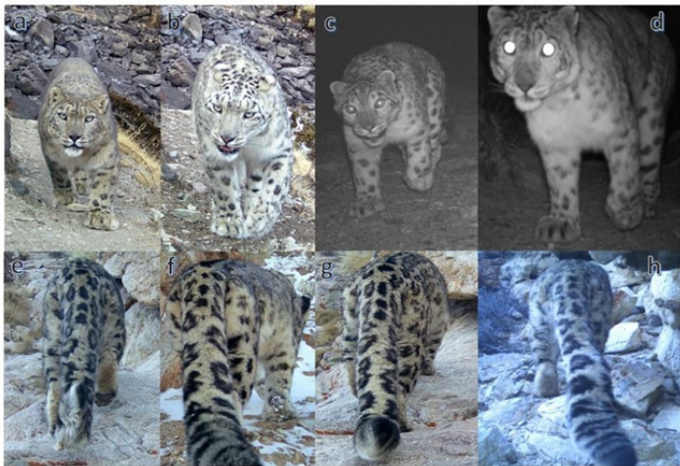
$N = ?$



Population and sample

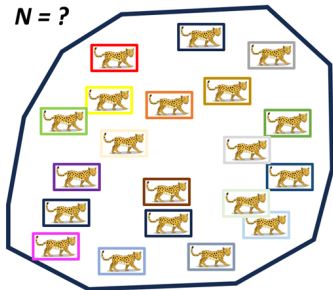


Snow leopards: Individual IDs

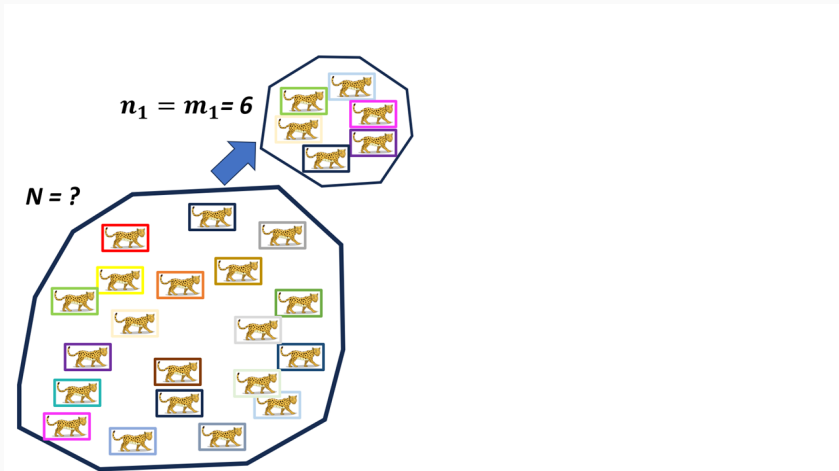


(Non-spatial) capture-recapture (CR)

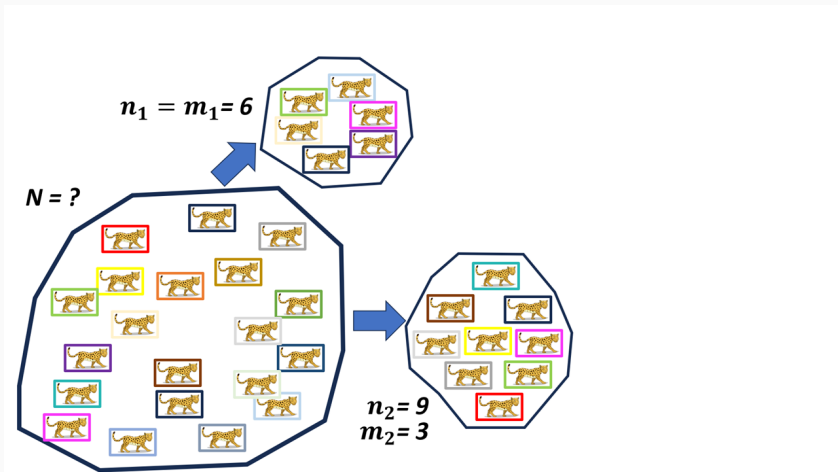
$N = ?$



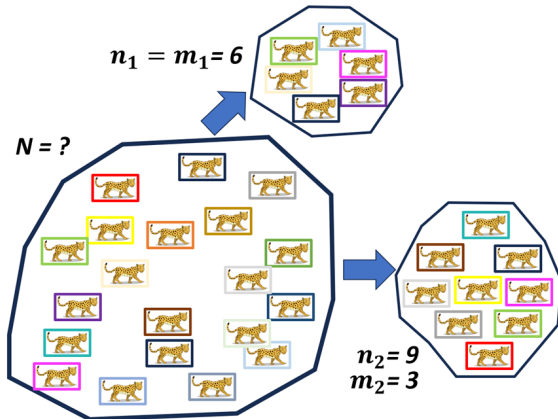
(Non-spatial) capture-recapture (CR)



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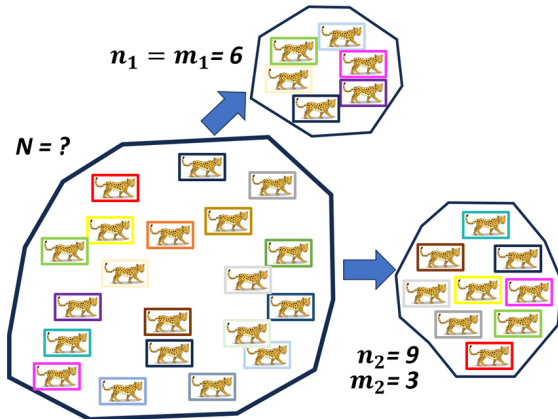
(Non-spatial) capture-recapture (CR)



$$\hat{p} = \frac{m_2}{m_1} = \frac{n_2}{N}$$

$$\hat{N} = \frac{n_2}{\hat{p}}$$

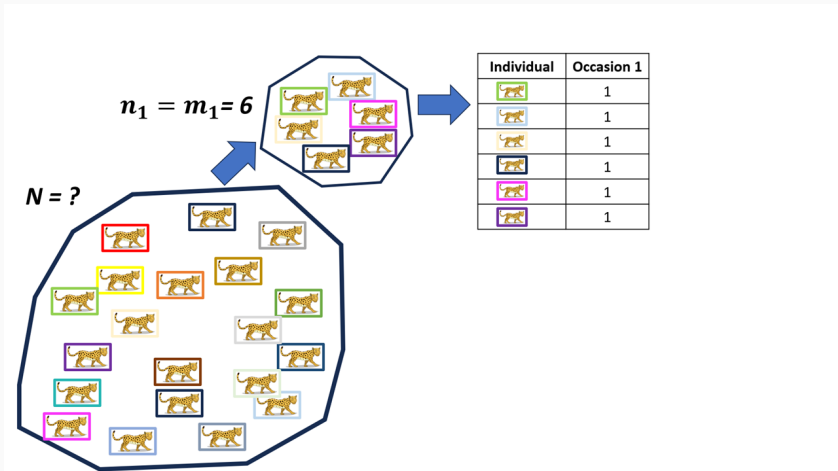
(Non-spatial) capture-recapture (CR)



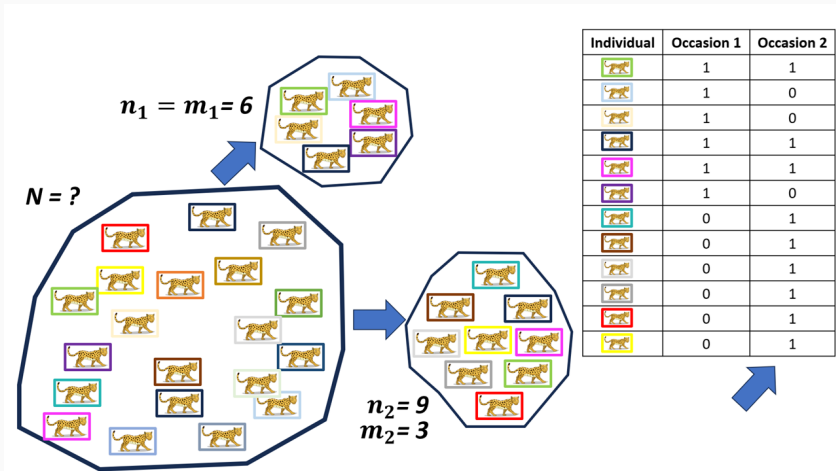
$$\hat{p} = \frac{m_2}{n_2} = \frac{3}{6} = 0.5$$

$$\hat{N} = \frac{n_1}{\hat{p}} = \frac{6}{0.5} = 12$$

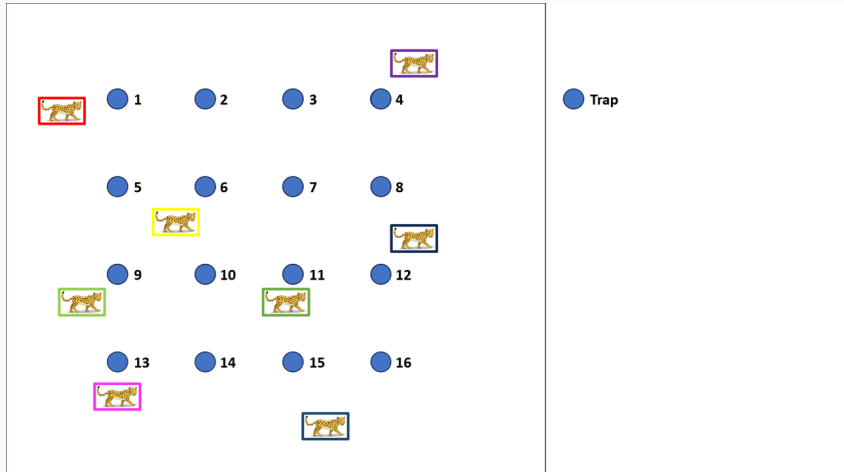
(Non-spatial) capture-recapture (CR): Capture histories



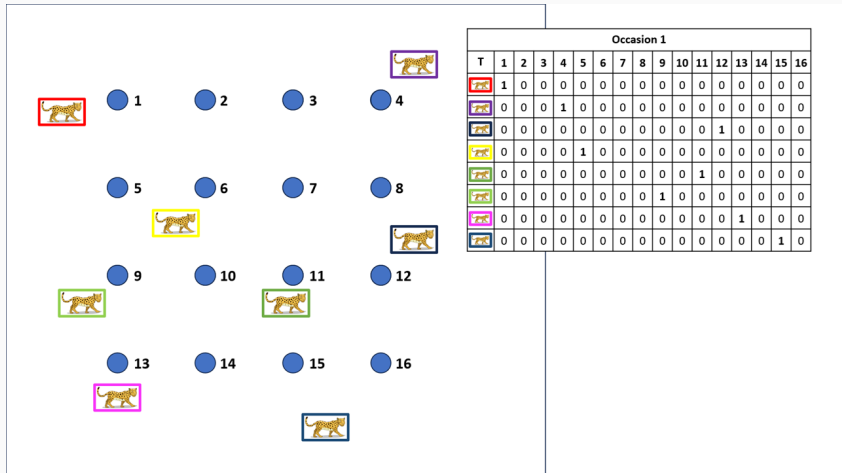
(Non-spatial) capture-recapture (CR): Capture histories



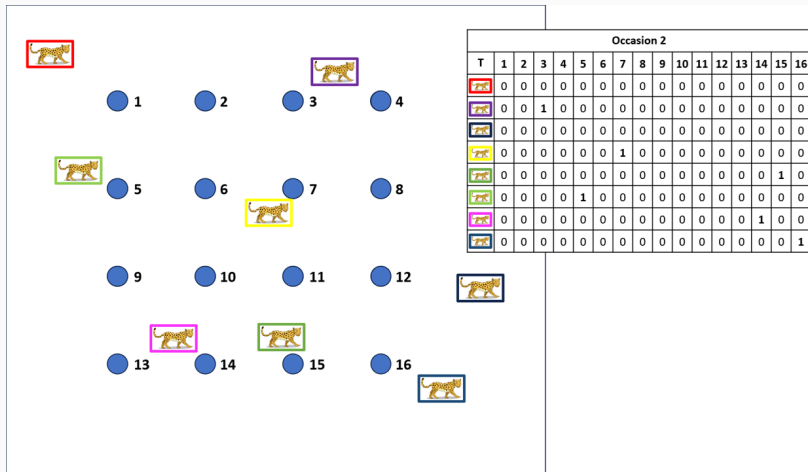
Spatial capture-recapture (SCR): Single/multi catch traps



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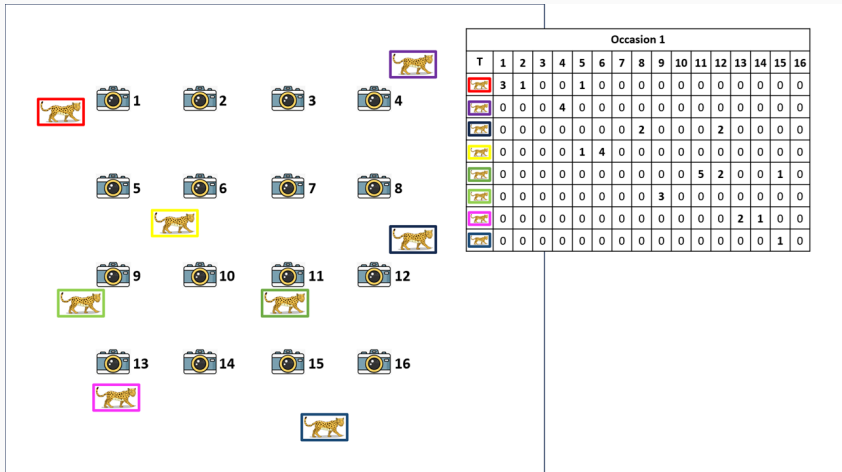
Spatial capture-recapture (SCR): Single/multi catch traps



Spatial capture-recapture (SCR): Single/multi catch traps

Occasion 4																
T	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Occasion 3																
T	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Occasion 2																
T	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Occasion 1																
T	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0
	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0
	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0
	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0
	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0
	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0
	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0

Spatial capture-recapture (SCR): Count traps

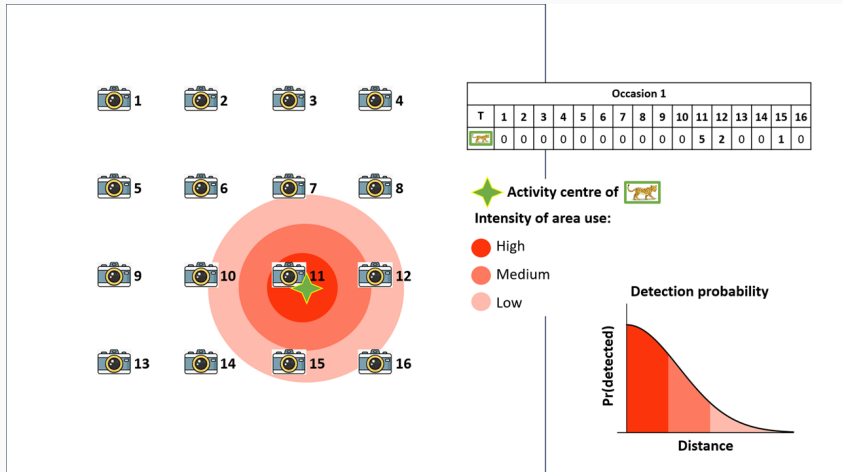


Spatial capture-recapture (SCR): Activity centre



Occasion 1																
T	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
	0	0	0	0	0	0	0	0	0	0	5	2	0	0	1	0

Spatial capture-recapture (SCR): Activity centre



Spatial capture-recapture (SCR): Detector types

Table 1: SCR (`secr`) detector types

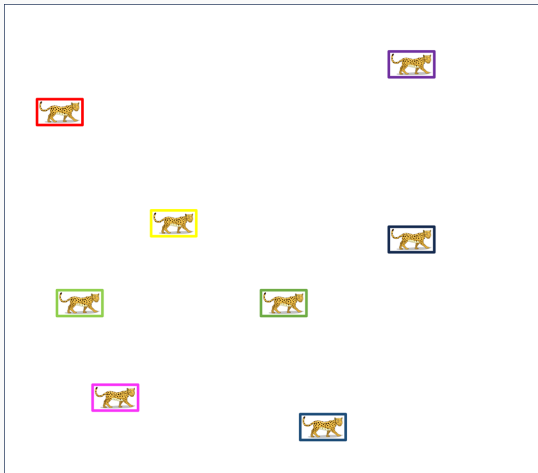
Category	Type (conceptual)	<code>secr</code> name
Point	Single-capture trap (1 animal at a time)	single
	Multi-capture trap (multiple animals)	multi
	Proximity detector (non-exclusive detection)	proximity
	Count detector (multiple detections per occasion)	count
	Binary proximity with cap of 1 detection	capped
	Acoustic / signal-based detector	signal
Area	Area search detector	polygon
	Area search with independent subdetectors	polygonX
Line	Linear search detector	transect
	Linear search with independent subdetectors	transectX

Spatial capture-recapture (SCR): What we know / observe



Occasion 1																
T	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
	3	1	0	0	1	0	0	0	0	0	0	0	0	0	0	0
	0	0	0	4	0	0	0	0	0	0	0	0	0	0	0	0
	0	0	0	0	0	0	0	2	0	0	0	2	0	0	0	0
	0	0	0	0	1	4	0	0	0	0	0	0	0	0	0	0
	0	0	0	0	0	0	0	0	0	0	5	2	0	0	1	0
	0	0	0	0	0	0	0	0	3	0	0	0	0	0	0	0
	0	0	0	0	0	0	0	0	0	0	0	0	2	1	0	0
	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0

SCR: What we want to know: abundance / density



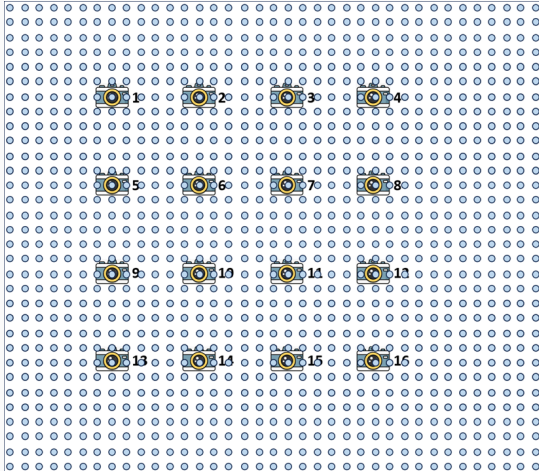
$\hat{N}$ Abundance

$\hat{D}$ Density

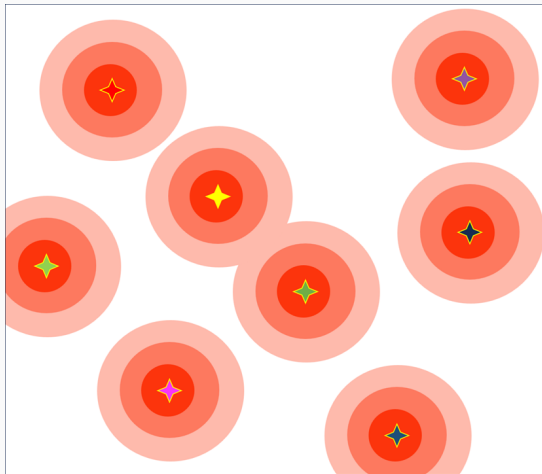
A Area

$$\hat{D} = \frac{\hat{N}}{A}$$

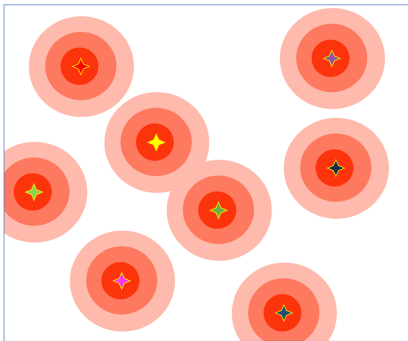
SCR: How secr does it: Mask and buffer



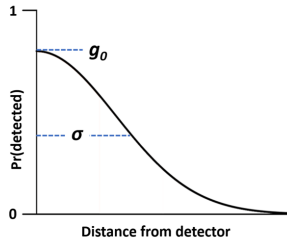
SCR: How secr does it: Activity centres



State model: Density D



Observation model



Spatial capture-recapture: Data

```
head(traps_df, 3)
```

	session	trapID	Latitude	Longitude	x	y	UTM.Zone	start_date
1	1	Camera8	30.23653	81.53307	551291	3345115	44R	17/12/2022
2	1	Camera11	30.20199	81.45040	543352	3341253	44R	20/12/2022
3	1	Camera14	30.16731	81.38865	537422	3337389	44R	21/12/2022
	end_date	effort	Topography	Altitude	Water	Note		
1	13/03/2023	86	Cliff	3978	1	Functional		
2	07/02/2023	49	Cliff	4061	0	Functional		
3	16/03/2023	85	Cliff	3959	0	Functional		

```
head(caps_df, 3)
```

	session	animalID	occasion	date	trapID	sex
1	1	SL1	1	02/16/2023	TPC37	Female
2	1	SL10	1	03/04/2023	TPC14	Female
3	1	SL10	1	05/13/2023	TPC14	Female

Spatial capture-recapture (SCR): Assumptions

Table 2: SCR Assumptions

#	Assumption	Description
1	Population closure	No births, deaths, immigration, or emigration occur during the sampling period.
2	Fixed activity centres	Each individual has a stable activity centre that does not change during the study.
3	Distance-dependent detection	Detection probability declines monotonically with distance from an individual's activity centre.
4	Independence of detections	Detection events are independent across individuals and sampling occasions (conditional on model parameters).

Spatial capture-recapture (SCR): Assumptions

Table 3: SCR Assumptions

#	Assumption	Description
5	Correct individual identification	All individuals are uniquely and correctly identified with no mis-classification errors.
6	Known detector locations	Trap or detector locations are fixed and known without error.
7	Detection homogeneity or modeled variation	Detection parameters are either constant across individuals or heterogeneity is explicitly modeled (e.g., via covariates).
8	Activity centre distribution	Activity centres arise from a defined spatial point process (often uniform or Poisson) within the state space.
9	Known sampling effort	Sampling effort, including detector operation and timing, is accurately recorded and incorporated into the model.

Spatial capture-recapture (SCR): Terminology

Table 4: SCR terminology and definitions (1)

Term	Definition
Capture history	A record of detections for each individual across detectors and sampling occasions.
Detector (trap)	A device or method used to detect individuals (e.g., camera trap, hair snare, acoustic recorder).
Detector array	The spatial configuration of all detectors used in the study.
Activity centre	The latent (unobserved) central location of an individual's home range, assumed fixed during the study.
State space	The spatial region that contains all possible activity centres of individuals in the population.
Mask	A discretized representation of the state space used in computation (grid of possible activity centre locations).
Buffer	An added margin around the detector array to ensure the state space includes all individuals that could be detected.

Spatial capture-recapture (SCR): Terminology

Table 5: SCR terminology and definitions (2)

Term	Definition
Session	A distinct sampling period with its own detector array, effort, and capture data (e.g., a study site or year).
Occasion	A within-session sampling period (e.g., day, night, week) during which detections are recorded.
Effort	Information on when and where detectors were active; used to correctly model detection probability.
Detection function	A function describing how detection probability declines with distance from the activity centre.
g_0	The probability of detecting an individual at its activity centre (i.e., distance = 0).
σ (sigma)	Spatial scale parameter governing how quickly detection probability declines with distance.
Density (D)	The number of individuals per unit area.

Spatial capture-recapture (SCR): Terminology

Table 6: SCR terminology and definitions (3)

Term	Definition
Encounter model	The statistical model describing how detections arise from the spatial process and detection process.
Spatial point process	The probabilistic model describing the distribution of activity centres across space.
Likelihood	The probability of observing the data given model parameters; used for parameter estimation.
Inhomogeneous density	A model where density varies across space as a function of environmental covariates.
Homogeneous density	A model assuming constant density across the state space.